

E010

Foundations of Programming (Python)

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A coffee shop that opens at 6 a.m. packs mugs 5 to a box. Its inventory script, written in a free text editor, runs `print(17 // 5, 17 % 5)`. What does it print?

- A. 3 0.4
- B. 2 3
- C. 3 2
- D. 3.4 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which Python expression evaluates to 3.5?

- A. `7 // 2`
- B. `int(7 / 2)`
- C. `7 / 2`
- D. `7 % 2`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A teaching assistant tests a loop in an IDE on a borrowed laptop: `for i in range(1, 5): print(i)`. Which numbers are printed?

- A. 0 1 2 3 4
- B. 1 2 3 4
- C. 2 3 4 5
- D. 1 2 3 4 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which `range()` call produces 0, 2, 4, 6, 8?

- A. `range(0, 8, 2)`
- B. `range(10, 0, -2)`
- C. `range(2, 10)`
- D. `range(0, 10, 2)`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A program running in a virtual environment on Python 3.12 has `a = [1, 2, 3]` followed by `b = a` and then `b.append(4)`. What does `print(a)` show?

- A. `[1, 2, 3, 4]`
- B. `[1, 2, 3]`
- C. An error, because `append` was not called on `a`.
- D. `[1, 2, 3]`, because assignment always copies a list.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Given `a = [1, 2, 3]`, which line creates `b` so that later changes to `b` do not affect `a`?

- A. `b == a`
- B. `None`; Python lists cannot be copied.
- C. `b = a`
- D. `b = a.copy()`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A script reads 500 course names from a CSV file. For one entry, `word = 'PYTHON'`. What does `word[1:4]` return?

- A. `YTH`
- B. `PYT`
- C. An error, because slicing only works on lists.
- D. `YTHO`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An instructor built a grading script for a class of 32 students. With `score = 85`, it runs: `if score >= 70: grade = 'C'; elif score >= 80: grade = 'B'; elif score >= 90: grade = 'A'`. What is grade?

- A. C, then B
- B. B
- C. C
- D. No grade, since 85 is not exactly 70, 80, or 90.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A shipping program asks how many boxes shipped Monday and Tuesday: `a = input()` and `b = input()`. The user types 5, then 3. What does `print(a + b)` show?

- A. 5 3
- B. 8
- C. A `TypeError`.
- D. 53

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In the same program (a and b from `input()`, user typed 5 and 3), which line prints 8?

- A. `print(a + b)`
- B. `print(int(a) + int(b))`
- C. `print(int(a + b))`
- D. `print(str(a) + str(b))`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A student's file `grades_v2.py` passes the IDE's linter with no warnings. It sums `[80, 90, 100]` into `total` and then computes `avg = total / 2`. It prints 135.0. What kind of error is this?

- A. A syntax error.
- B. A logic error.
- C. A runtime error.
- D. No error, since the linter found none.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A grocery app, updated every Monday by the store manager, has `prices = {'apple': 0.5, 'pear': 0.75}`. Customers sometimes ask about plums, which are not listed. Which expression returns 0 instead of crashing?

- A. `prices.get('plum', 0)`
- B. `prices.get(0, 'plum')`
- C. `prices[0]`
- D. `prices['plum']`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A coffee shop ~~that opens at 6 a.m.~~ packs mugs 5 to a box. Its inventory script, ~~written in a free text editor,~~ runs `print(17 // 5, 17 % 5)`. What does it print?

Target word: output of // and % **Decides:** // gives the whole quotient; % gives the remainder.

A	Adjacent category	Reports the fractional part instead of the integer remainder.
B	Inverted operation	Swaps the quotient and remainder.
C	Correct	$17 // 5 = 3$ full boxes; $17 \% 5 = 2$ left over.
D	Confusion set	Treats // as regular division (/).

2. Answer C Concept 1 · transfer

Which Python expression evaluates to 3.5?

Target word: pick the operator — reverse direction **Decides:** Only / keeps the decimal.

A	Confusion set	Floor division returns 3.
B	Dropped element	<code>int()</code> drops the decimal, giving 3.
C	Correct	True division returns 3.5.
D	Adjacent category	The remainder is 1.

3. Answer B Concept 2

A teaching assistant tests a loop ~~in an IDE on a borrowed laptop:~~ `for i in range(1, 5): print(i)`. Which numbers are printed?

Target word: values produced by range **Decides:** Start is included; stop (5) is excluded.

A	Wrong stem number	Ignores the start value of 1 and uses the default 0.
B	Correct	range stops before 5.
C	Inverted operation	Excludes the start and includes the stop — backward.
D	Confusion set	Includes the stop value.

4. Answer D Concept 2 · transfer

Which `range()` call produces 0, 2, 4, 6, 8?

Target word: build the range — reverse of reading one **Decides:** Stop must be past 8; step is 2.

A	Dropped element	Stops before 8, so it ends at 6.
B	Inverted operation	Counts down: 10, 8, 6, 4, 2.
C	Confusion set	Reads the 2 as a step, but it is the start; this gives 2 through 9.
D	Correct	Starts at 0, steps by 2, stops before 10.

5. Answer A Concept 3

A program ~~running in a virtual environment on Python 3.12~~ has `a = [1, 2, 3]` followed by `b = a` and then `b.append(4)`. What does `print(a)` show?

Target word: effect of aliasing **Decides:** `a` and `b` refer to the same list.

A	Correct	Appending through <code>b</code> changes the one shared list.
B	Confusion set	Assumes <code>b = a</code> made a copy.
C	Dropped element	Drops the fact that <code>b</code> and <code>a</code> are the same object.
D	Absolute language	Assignment never copies; it creates another name.

6. Answer D Concept 3 · transfer

Given `a = [1, 2, 3]`, which line creates `b` so that later changes to `b` do not affect `a`?

Target word: make an independent copy **Decides:** A new list object is required.

A	Confusion set	Compares the lists; it assigns nothing.
B	Absolute language	<code>copy()</code> and slicing both copy lists.
C	Inverted operation	Creates an alias to the same list.
D	Correct	<code>copy()</code> creates a new list with the same items.

7. Answer A Concept 4

A script ~~reads 500 course names from a CSV file. For one entry,~~ `word = 'PYTHON'`. What does `word[1:4]` return?

Target word: slice result **Decides:** Index 1 is Y; the slice stops before index 4.

A	Correct	Indexes 1, 2, 3 are Y, T, H.
B	Confusion set	Counts positions from 1 instead of 0.
C	Absolute language	Strings support slicing too.
D	Dropped element	Drops the rule that the end index is excluded.

8. Answer C Concept 5

An instructor built a grading script ~~for a class of 32 students~~. With `score = 85`, it runs: `if score >= 70: grade = 'C'; elif score >= 80: grade = 'B'; elif score >= 90: grade = 'A'`. What is `grade`?

Target word: which branch runs **Decides:** `85 >= 70` is true first, so later branches are skipped.

A	Dropped element	Drops the rule that <code>elif</code> stops after one true branch.
B	Confusion set	Picks the best-fitting branch, but Python stops at the first true one.
C	Correct	Only the first true branch runs.
D	Inverted operation	Reads <code>>=</code> as <code>==</code> .

9. Answer D Concept 6

A shipping program asks ~~how many boxes shipped Monday and Tuesday~~: `a = input()` and `b = input()`. The user types 5, then 3. What does `print(a + b)` show?

Target word: result of `+` on inputs **Decides:** `input()` returns strings, so `+` concatenates.

A	Dropped element	That is <code>print(a, b)</code> , which adds a space separator.
B	Confusion set	Treats the inputs as numbers.
C	Adjacent category	String <code>+</code> string is valid; the error comes from mixing a string and a number.
D	Correct	<code>'5' + '3'</code> joins the text.

10. Answer B Concept 6 · transfer

In the same program (a and b from input(), user typed 5 and 3), which line prints 8?

Target word: fix the conversion **Decides:** Each value must become a number before adding.

A	Dropped element	Leaves out the conversion entirely.
B	Correct	Converts each string to an integer, then adds.
C	Confusion set	Concatenates first, then converts '53' to 53.
D	Inverted operation	Keeps them as text, printing 53.

11. Answer B Concept 7

A student's file `grades_v2.py` passes the IDE's linter with no warnings. It sums [80, 90, 100] into total and then computes `avg = total / 2`. It prints 135.0. What kind of error is this?

Target word: classify the error **Decides:** Runs without crashing but gives a wrong answer.

A	Confusion set	Syntax errors prevent the program from running at all.
B	Correct	Dividing by 2 instead of 3 produces a wrong result without any error message.
C	Adjacent category	Runtime errors crash the program; this one finished.
D	Camouflage	Linters catch style and syntax issues, not wrong math.

12. Answer A Concept 8

A grocery app, ~~updated every Monday by the store manager~~, has prices = {'apple': 0.5, 'pear': 0.75}. Customers sometimes ask about plums, which are not listed. Which expression returns 0 instead of crashing?

Target word: safe dictionary lookup **Decides:** The key may be missing.

A	Correct	get returns the default when the key is missing.
B	Inverted operation	Swaps the key and default, returning 'plum'.
C	Confusion set	Dictionaries use keys, not positions; this raises KeyError.
D	Dropped element	Raises KeyError for a missing key.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	12	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Division operators	/ always returns a float; // returns the floor (whole-number) quotient; % returns the remainder.	Expecting / to drop the decimal or mixing up // and %.	An expression with /, //, or % on two integers.
2	range() stops before the end value	range(start, stop, step) includes start and excludes stop.	Including the stop value or assuming it starts at 1.	A for loop over range with two or three arguments.
3	List aliasing	b = a makes both names point to the same list; changes through either show in both. Use a.copy() or a[:] for a separate list.	Thinking assignment copies a list.	One list assigned to a second name, then modified.
4	Zero-based indexing and slicing	Indexes start at 0; a slice s[i:j] includes i and stops before j.	Counting from 1 or including the end index.	A string or list slice with two numbers.
5	if / elif runs only the first true branch	Python checks conditions top to bottom and runs only the first branch that is true.	Expecting the “best” match or every true branch to run.	Overlapping conditions like >= 70, >= 80, >= 90 in that order.
6	input() returns a string	input() always returns text; convert with int() or float() before doing math.	Expecting + on two inputs to add numbers.	Two input() values combined with +.
7	Syntax, runtime, and logic errors	Syntax errors stop the program before it runs; runtime errors crash it while running; logic errors run fine but give wrong results.	Assuming a program with no error message is correct.	Code that runs cleanly but prints a wrong answer.
8	Dictionary lookup	d[key] raises KeyError for a missing key; d.get(key, default) returns the default instead.	Indexing a dictionary by position like a list.	Looking up a key that may not exist.

Course scope (public description): Foundations of Programming (Python) introduces students to the fundamental principles of programming using Python. This course is designed to equip students with the essential skills needed to write, troubleshoot, and debug simple Python programs. Students will explore the syntax and structure of Python, develop an understanding of variables, loops, and conditional statements, and learn to work within different development environments, including integrated development environments (IDEs) and text editors. By the end of the course, students will be able to create functional Python programs, manipulate data structures, and apply debugging techniques across various platforms. There are no prerequisites for this course.